

"Zhuangzi", Fit for Emperors and Kings
The Death of Primal Chaos

南海之帝為儵 北海之帝為忽 中央之帝為渾沌

儵与忽 時相与遇於渾沌之地

渾沌待之甚善

儵与忽謀報渾沌之德曰

人皆有七竅 以視聽食息

此独無有

嘗試鑿之

日鑿一竅 七日而渾沌死

『莊子』 應帝王篇



Signatures of quantum chaos in quantum spin systems imaginary-time evolution and operator growth

Hal Tasaki

*H. Tasaki, presentation at the 2025 (hot) Summer Meeting of the Physical Society of Japan and a paper in preparation,
N. Shiraishi and H. Tasaki, "The $S = 1/2$ XY and XYZ models on the two or higher dimensional hypercubic lattice do not
possess nontrivial local conserved quantities", arXiv:2412.18504*

webinar@YouTube, October 2025

motivation
definitions and main results
main ideas of the proof
summary and discussion



mathematical studies of quantum many-body systems

exact solutions

free fermion, Bethe ansatz, Yang-Baxter relation ...

integrable models

rigorous, general theorems

integrable models

non-integrable models

there are properties/phenomena (quantum chaos, thermalization, ETH = energy eigenstate thermalization hypothesis, ...) that are expected to take place only in (non-random) non-integrable systems

mathematical results that exclusively apply to (non-random) non-integrable systems

??

early results that proved a concrete quantum model (with short-range non-random interactions) exhibits a behavior that is never observed in integrable models

Bouch (2015) singularity in the imaginary-time evolution in a two-dimensional quantum spin system

$$\hat{H}_B := \sum_{j,k \in \mathbb{Z}} \{ \hat{X}_{(j,k)} \hat{Z}_{(j+1,k)} + \hat{Z}_{(j,k)} \hat{X}_{(j,k+1)} \}$$



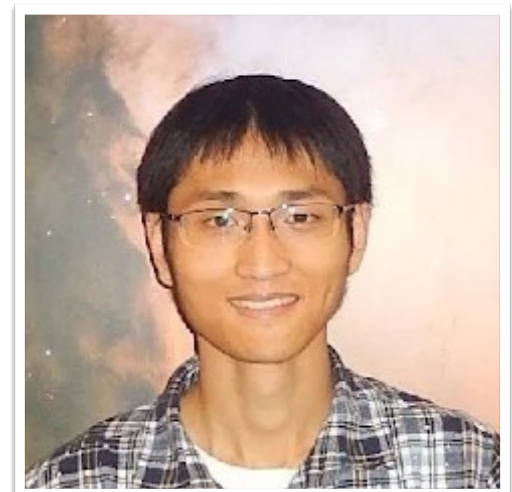
Gabriel Bouch

Shiraishi (2019) absence of nontrivial local conserved quantities in the $S = 1/2$ XYZ-h spin chain



Naoto Shiraishi

Cao (2021) “quantum chaotic” behavior of the moment in the one and two-dimensional quantum Ising models



Xiangyu Cao

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$S = \frac{1}{2}$ quantum spin model on $\mathbb{Z}^d$

d -dimensional hypercubic lattice $\mathbb{Z}^d$

$\hat{X}_u, \hat{Y}_u, \hat{Z}_u$ copies of Pauli operators $\hat{X}, \hat{Y}, \hat{Z}$ at site $u \in \mathbb{Z}^d$

algebra of local operators

$$\mathfrak{A}_{\text{loc}} = \{\text{polynomials of } \hat{X}_u, \hat{Y}_u, \hat{Z}_u \text{ with } u \in \mathbb{Z}^d\}$$

(formal) Hamiltonian on $\mathbb{Z}^d$

$$\begin{aligned} \hat{H} = & -\frac{1}{2} \sum_{\substack{u, v \in \mathbb{Z}^d \\ (|u-v|=1)}} \{ J_X \hat{X}_u \hat{X}_v + J_Y \hat{Y}_u \hat{Y}_v + J_Z \hat{Z}_u \hat{Z}_v \} \\ & - \sum_{u \in \mathbb{Z}^d} \{ h_X \hat{X}_u + h_Y \hat{Y}_u + h_Z \hat{Z}_u \} \\ & J_X, J_Y, J_Z, h_X, h_Y, h_Z \in \mathbb{R} \end{aligned}$$

$\hat{H}_\Lambda \in \mathfrak{A}_{\text{loc}}$ the restriction of $\hat{H}$ onto a finite region $\Lambda \subset \mathbb{Z}^d$

side remark: absence of nontrivial local conserved quantities

$$\hat{H} = -\frac{1}{2} \sum_{\substack{u,v \in \mathbb{Z}^d \\ (|u-v|=1)}} \{J_X \hat{X}_u \hat{X}_v + J_Y \hat{Y}_u \hat{Y}_v + J_Z \hat{Z}_u \hat{Z}_v\} - \sum_{u \in \mathbb{Z}^d} \{h_X \hat{X}_u + h_Y \hat{Y}_u + h_Z \hat{Z}_u\}$$
$$S = \frac{1}{2}$$

Theorem: in all models with $d \geq 2$ except for the classical Ising model, there exist no nontrivial local conserved quantities

Chiba 2024, Shiraishi, Tasaki 2024

classical Ising model

$J_X = J_Y = h_X = h_Y = 0, J_Y = J_Z = h_Y = h_Z = 0$, **or** $J_Z = J_X = h_Z = h_X = 0$

strong indication that the models are non-integrable
extensions of Shiraishi's groundbreaking work in 2019

extremely rapid and essential progress in this direction!

time-evolution for the infinite system

follows from the Lieb-Robinson bound

Theorem for any $\hat{A} \in \mathfrak{A}_{\text{loc}}$ and $t \in \mathbb{R}$, the infinite volume limit $\hat{A}(t) = \lim_{\Lambda \uparrow \mathbb{Z}^d} e^{i\hat{H}_\Lambda t} \hat{A} e^{-i\hat{H}_\Lambda t} \in \mathfrak{A} := \overline{\mathfrak{A}_{\text{loc}}}$ exists (convergence in the operator norm)

Theorem (Araki, 1969) in $d = 1$, the same limit $\hat{A}(t)$ exists for any $\hat{A} \in \mathfrak{A}_{\text{loc}}$ and $t \in \mathbb{C}$

what about systems in $d \geq 2$?

the limit exists for any $\hat{A} \in \mathfrak{A}_{\text{loc}}$ and $t \in \mathbb{C}$ s.t. $|t| \leq r_0$

Theorem (Bouch, 2015) there is a model with $d = 2$ and $\hat{A} \in \mathfrak{A}_{\text{loc}}$, where the limit $\hat{A}(i\beta)$ does not converge in $\mathfrak{A}$ (in the operator norm) for $\beta \in \mathbb{R}$ with sufficiently large $|\beta|$

$$\hat{H}_B = \sum_{(j,k) \in \mathbb{Z}^2} \{ \hat{X}_{(j,k)} \hat{Z}_{(j+1,k)} + \hat{Z}_{(j,k)} \hat{X}_{(j,k+1)} \}$$

imaginary-time evolution and characterization of quantum chaos

Avdoshkin, Dymarsky 2020

$$\hat{A}(i\beta) = \lim_{\Lambda \uparrow \mathbb{Z}^d} e^{-\beta \hat{H}_\Lambda} \hat{A} e^{\beta \hat{H}_\Lambda}$$

in an integrable system, $\hat{A}(i\beta)$ with $\beta \in \mathbb{R}$ grows (in space) at most polynomially in $|\beta|$

in a chaotic system, $\hat{A}(i\beta)$ with $\beta \in \mathbb{R}$ grows (in space) in the fastest possible manner

$d = 1$ double exponential in $|\beta|$

$d \geq 2$ reaches infinity at a finite $|\beta|$

no Lieb-Robinson!

related in an essential manner to the universal operator growth hypothesis

main results on complex-time evolution

$$\hat{H} = -\frac{1}{2} \sum_{\substack{u,v \in \mathbb{Z}^d \\ (|u-v|=1)}} \{J_X \hat{X}_u \hat{X}_v + J_Y \hat{Y}_u \hat{Y}_v + J_Z \hat{Z}_u \hat{Z}_v\} - \sum_{u \in \mathbb{Z}^d} \{h_X \hat{X}_u + h_Y \hat{Y}_u + h_Z \hat{Z}_u\} \quad S = \frac{1}{2}$$

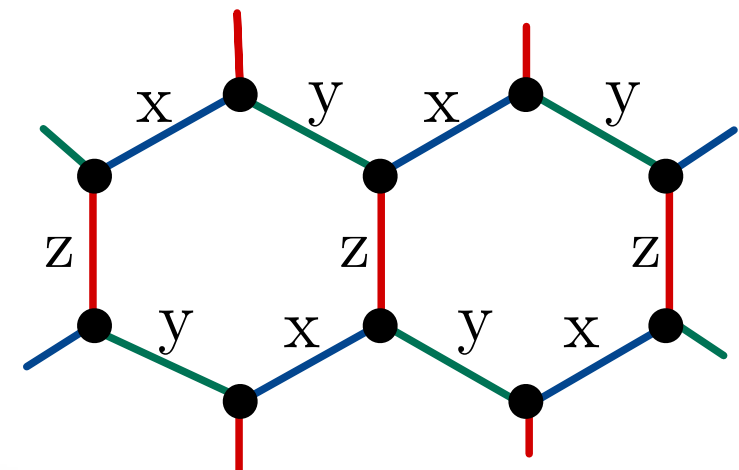
Theorem: in all models with $d \geq 2$ except for the classical Ising model, $\hat{A}(i\beta)$ with some $\hat{A} \in \mathfrak{A}_{\text{loc}}$ does not converge for $\beta \in \mathbb{R}$ with sufficiently large $|\beta|$

Shiraishi, Tasaki 2024, Tasaki unpublished

this extends Bouch's theorem to "standard" spin models

integrable case

$$\hat{H}_{\text{Kitaev}} = \sum_{\{u,v\} \in \mathcal{B}_x} \hat{X}_u \hat{X}_v + \sum_{\{u,v\} \in \mathcal{B}_y} \hat{Y}_u \hat{Y}_v + \sum_{\{u,v\} \in \mathcal{B}_z} \hat{Z}_u \hat{Z}_v$$



Theorem: in the Kitaev honeycomb model, the limit $\hat{A}(t)$ exists for any $\hat{A} \in \mathfrak{A}_{\text{loc}}$ and any $t \in \mathbb{C}$

Tasaki unpublished

main results on complex-time evolution

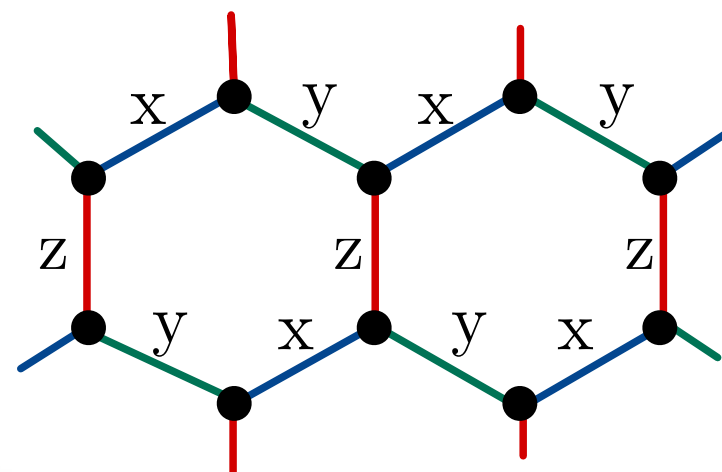
$$\hat{H} = -\frac{1}{2} \sum_{\substack{u,v \in \mathbb{Z}^d \\ (|u-v|=1)}} \{J_X \hat{X}_u \hat{X}_v + J_Y \hat{Y}_u \hat{Y}_v + J_Z \hat{Z}_u \hat{Z}_v\} - \sum_{u \in \mathbb{Z}^d} \{h_X \hat{X}_u + h_Y \hat{Y}_u + h_Z \hat{Z}_u\} \quad S = \frac{1}{2}$$

Theorem: in all models with $d \geq 2$ except for the classical Ising model, $\hat{A}(i\beta)$ with some $\hat{A} \in \mathfrak{A}_{\text{loc}}$ does not converge for $\beta \in \mathbb{R}$ with sufficiently large $|\beta|$

this extends consistent with the conjecture by Tasaki unpublished models
Avdoshkin and Dymarsky

integrable case

$$\hat{H}_{\text{Kitaev}} = \sum_{\{u,v\} \in \mathcal{B}_x} \hat{X}_u \hat{X}_v + \sum_{\{u,v\} \in \mathcal{B}_y} \hat{Y}_u \hat{Y}_v + \sum_{\{u,v\} \in \mathcal{B}_z} \hat{Z}_u \hat{Z}_v$$



Theorem: in the Kitaev honeycomb model, the limit $\hat{A}(t)$ exists for any $\hat{A} \in \mathfrak{A}_{\text{loc}}$ and any $t \in \mathbb{C}$

operator growth

normalized Hilbert-Schmidt inner product

$$\langle \hat{A}, \hat{B} \rangle_{\text{NHS}} := \text{Tr}_{\mathcal{H}_\Lambda} [\hat{A}^\dagger \hat{B}] / \text{Tr}_{\mathcal{H}_\Lambda} [\hat{1}]$$

$$\hat{A}, \hat{B} \in \mathfrak{A}_{\text{loc}}$$

$\mathcal{H}_\Lambda$ Hilbert space on Λ

Liouvillian (or generator of time-evolution)

$$\delta(\hat{A}) := i [\hat{H}_\Lambda, \hat{A}]$$

for sufficiently
large $\Lambda \subset \mathbb{Z}^d$

initial operator $\hat{A} \in \mathfrak{A}_{\text{loc}}$ s.t. $\hat{A}^\dagger = \hat{A}$ and $\langle \hat{A}, \hat{A} \rangle_{\text{NHS}} = 1$

$$\hat{A}, \delta(\hat{A}), \delta^2(\hat{A}), \delta^3(\hat{A}), \dots$$

$$\hat{O}_0 = \hat{A}$$



Gram-Schmidt

$$\hat{O}_n = \delta^n(\hat{A}_0) + \sum_{j=0}^{n-1} \alpha_j^{(n)} \delta^j(\hat{A}_0)$$

with $\langle \hat{O}_n, \hat{O}_j \rangle_{\text{NHS}} = 0$ for all $j = 0, \dots, n-1$

$$\hat{O}_0, \hat{O}_1, \hat{O}_2, \hat{O}_3, \dots$$

$\hat{O}_n$ the operator that appears “for the first time” in the n -th recursion $\delta^n(\hat{A})$

Lanczos coefficient

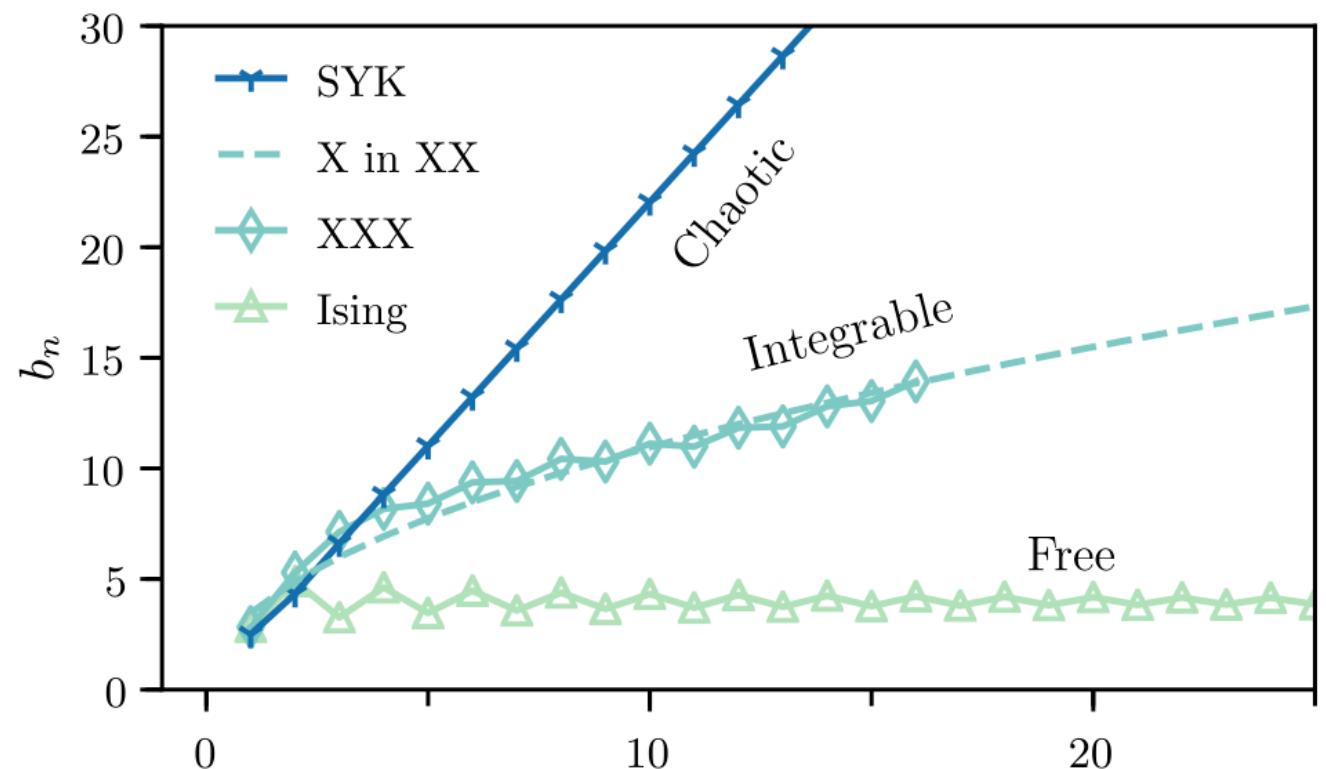
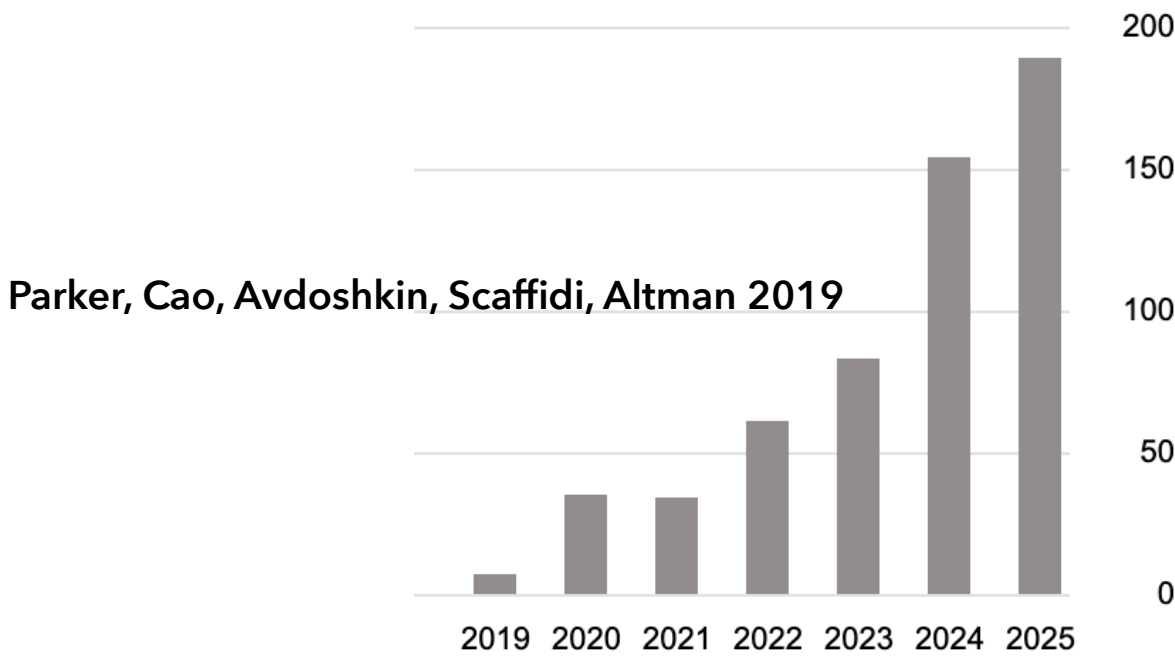
$$b_n := \sqrt{\frac{\langle \hat{O}_n, \hat{O}_n \rangle_{\text{NHS}}}{\langle \hat{O}_{n-1}, \hat{O}_{n-1} \rangle_{\text{NHS}}}}$$

universal operator growth hypothesis

Parker, Cao, Avdoshkin, Scaffidi, Altman 2019
Nandy, Matsoukas-Roubeas, Martinez-Azcona, Dymarsky, del Campo 2024

the growth of the Lanczos coefficients b_n captures the essential feature of the quantum dynamics

$$b_n \sim \begin{cases} \text{const} & \text{free models} \\ \text{const } n^\delta \quad (0 < \delta < 1) & \text{interacting integrable models} \\ \begin{cases} \text{const } n & d \geq 2 \\ \text{const } \frac{n}{\log n} & d = 1 \end{cases} & \text{chaotic models} \end{cases}$$



main result on operator growth

$$\hat{H} = -\frac{1}{2} \sum_{\substack{u,v \in \mathbb{Z}^d \\ (|u-v|=1)}} \{J_X \hat{X}_u \hat{X}_v + J_Y \hat{Y}_u \hat{Y}_v + J_Z \hat{Z}_u \hat{Z}_v\} - \sum_{u \in \mathbb{Z}^d} \{h_X \hat{X}_u + h_Y \hat{Y}_u + h_Z \hat{Z}_u\}$$
$$S = \frac{1}{2}$$

Theorem: in all models with $d \geq 2$ except for the classical Ising model, there exist $\hat{A} \in \mathfrak{A}_{\text{loc}}$, $\alpha > 0$, and an infinite set $G \subset \mathbb{N}$ such that $b_1 b_2 \cdots b_n \geq \alpha^n n!$ for any $n \in G$

Shiraishi, Tasaki 2024, Tasaki unpublished

essentially shows $b_n \gtrsim \alpha n$, the behavior expected in systems exhibiting quantum chaos!

we prove lower bounds only for the product $b_1 b_2 \cdots b_n$

$G \subset \mathbb{N}$ is an infinite set, but it is extremely sparse

the divergence of $\hat{A}(i\beta)$ follows from the above theorem

proof for the quantum Ising model relies heavily on the idea in Cao's 2021 paper

The emperor of the South Sea was called Shu [Brief], the emperor of the North Sea was called Hu [Sudden], and the emperor of the central region was called Hun-tun [Chaos]. Shu and Hu from time to time came together for a meeting in the territory of Hun-tun, and Hun-tun treated them very generously. Shu and Hu discussed how they could repay his kindness. "All men," they said, "have seven openings so they can see, hear, eat, and breathe. But Hun-tun alone doesn't have any. Let's try boring him some!"

Every day they bored another hole, and on the seventh day Hun-tun died.

"Zhuangzi", Chapter 7 Fit for Emperors and Kings

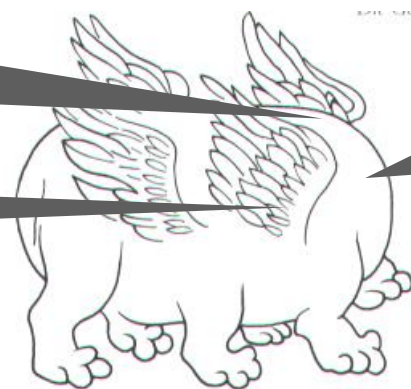
<https://terebess.hu/english/chuangtzu.html>

so far, we have bored only three (not very critical) holes

linear growth of b_n

no conserved quantities

singularity at imaginary time



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$$\hat{H} = -\frac{1}{2} \sum_{\substack{u,v \in \mathbb{Z}^2 \\ (|u-v|=1)}} \{J_X \hat{X}_u \hat{X}_v + J_Y \hat{Y}_u \hat{Y}_v + J_Z \hat{Z}_u \hat{Z}_v\} - \sum_{u \in \mathbb{Z}^2} \{h_X \hat{X}_u + h_Y \hat{Y}_u + h_Z \hat{Z}_u\}$$

$$S = \frac{1}{2} \quad J_X \neq 0, J_Y \neq 0$$

operator growth from $\hat{X}_o$ **Liouvillian**

$$\hat{X}_o, \delta(\hat{X}_o), \delta^2(\hat{X}_o), \delta^3(\hat{X}_o), \dots$$

$$\delta(\hat{A}) := i [\hat{H}, \hat{A}]$$



Gram-Schmidt

$$\hat{O}_0, \hat{O}_1, \hat{O}_2, \hat{O}_3, \dots$$

$\mathcal{P}$ the set of all products of Pauli operators on $\mathbb{Z}^d$

expand in products $\delta^n(\hat{X}_o) = \sum_{B \in \mathcal{P}} c_B B$

if there is B that does not appear in $\delta^j(\hat{X}_o)$ with $j = 0, 1, \dots, n-1$, then we have

$$\hat{O}_n = c_B B + \sum_{B' \neq B} \tilde{c}_{B'} B'$$

$$b_1 b_2 \cdots b_n = \|\hat{O}_n\|_{\text{NHS}} \geq |c_B|$$

$$\|\hat{A}\|_{\text{NHS}} := \sqrt{\langle \hat{A}, \hat{A} \rangle_{\text{NHS}}}$$

$$\langle A, B \rangle_{\text{NHS}} = \delta_{A,B}$$

$$A, B \in \mathcal{P}$$

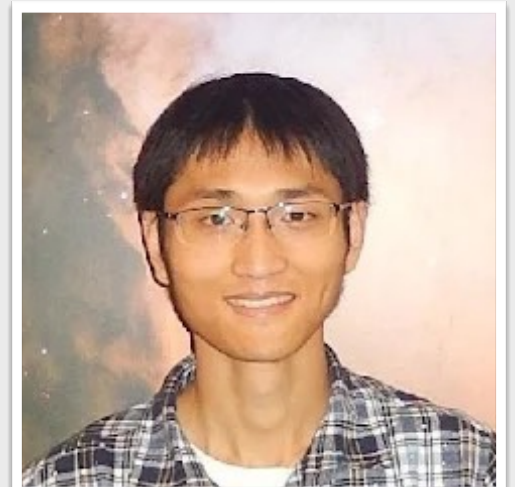
we shall look for such B with large $|c_B|$

$$\hat{H} = -\frac{1}{2} \sum_{\substack{u,v \in \mathbb{Z}^2 \\ (|u-v|=1)}} \{J_X \hat{X}_u \hat{X}_v + J_Y \hat{Y}_u \hat{Y}_v + J_Z \hat{Z}_u \hat{Z}_v\} - \sum_{u \in \mathbb{Z}^2} \{h_X \hat{X}_u + h_Y \hat{Y}_u + h_Z \hat{Z}_u\}$$

$$S = \frac{1}{2} \quad J_X \neq 0, J_Y \neq 0$$

operator ground state treatment of the quantum Ising model

$\hat{X}_o, \delta(\hat{X}_o), \delta^2(\hat{X}_o)$
 $J_X = J_Y = 0, J_Z \neq 0, h_X \neq 0$
requires extra care with the idea from Cao 2021



Xiangyu Cao

$\hat{O}_0, \hat{O}_1, \hat{O}_2, \hat{O}_3$

expand in products $\delta^n(\hat{X}_o) = \sum_{B \in \mathcal{P}} c_B B$

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operator growth from $\hat{X}_o$ **Liouvillian**

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$$\langle A, B \rangle_{\text{NHS}} = \delta_{A,B}$$

$$A, B \in \mathcal{P}$$

we shall look for such B with large $|c_B|$

some examples of B and c_B

$$\dots \boxed{\dots} \longleftrightarrow [\hat{X}_u \hat{X}_v, \dots \dots] \qquad \dots \boxed{\dots} \longleftrightarrow [\hat{Y}_u \hat{Y}_v, \dots \dots]$$

the simplest construction

$$\boxed{X} \rightarrow Z \boxed{Y} \rightarrow ZZ \boxed{X} \rightarrow \overbrace{ZZZ \dots ZZ}^n X = B$$

$c_B = (-4J_X J_Y)^{n/2}$

grown in a unique manner

appears for the first time in $\delta^n(\hat{X}_o)$

a better strategy

$$\overbrace{ZZZ \dots ZZ}^{n/2} X \rightarrow Z \boxed{Z} Z \dots Z \boxed{Z} X \rightarrow \begin{matrix} X & X \\ X Y X & \dots & X Y X \\ Y & Y & Y \end{matrix} = B$$

grow this

take $n/2$ commutation relations in an arbitrary order

grown in at least $(n/2)!$ ways (in fact $(n-1)!!$ ways)

$$b_1 b_2 \dots b_n \geq |c_B| \geq (4|J_X J_Y|)^{n/2} \left(\frac{n}{2}\right)! \qquad b_n \gtrsim \alpha n^{1/2}$$

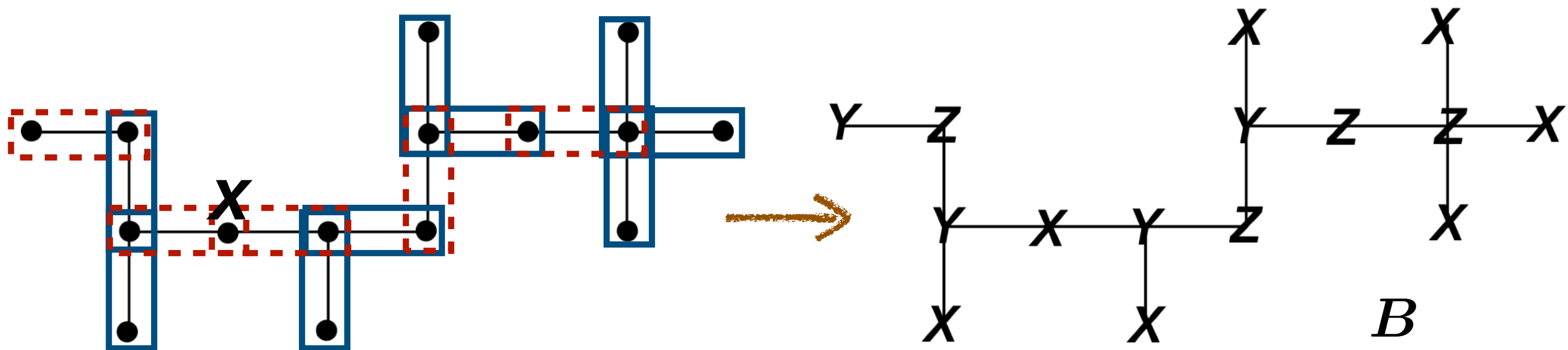
general construction on a rooted tree

$$T \subset \mathbb{Z}^2, |T| = n + 1, T \ni o$$

$$\tilde{T} = \{ \{u, v\} \mid u, v \in T, |u - v| = 1 \}$$

assume $(T, \tilde{T})$ is connected and contains no loops

we start from $\hat{X}_o$ and grow B supported on T



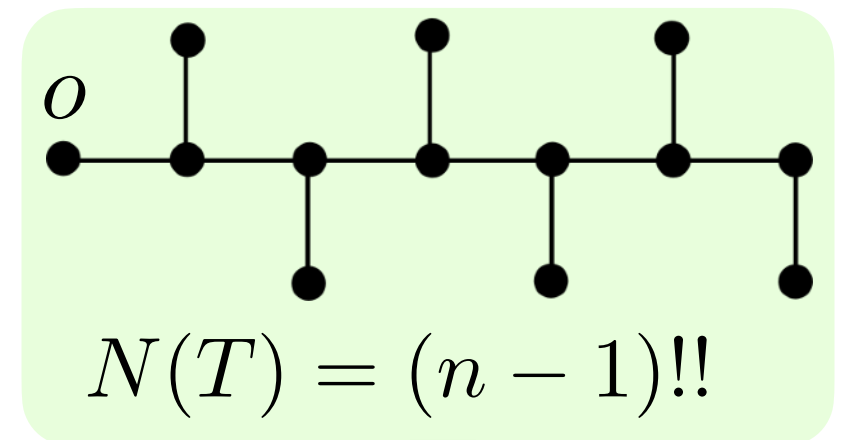
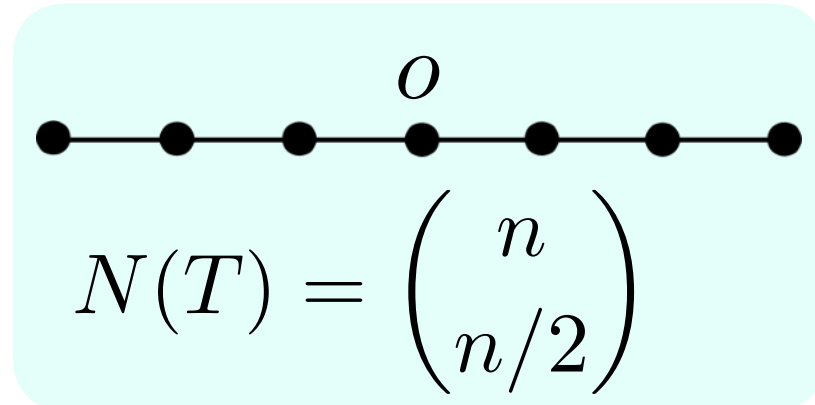
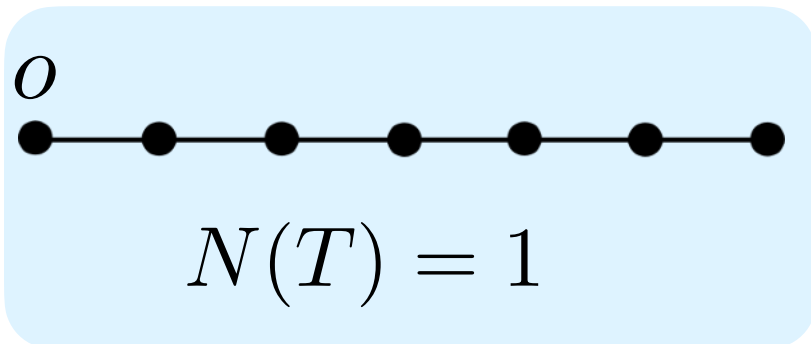
then $|c_B| \geq (2 \min\{|J_X|, |J_Y|\})^n N(T)$

$N(T)$ the number of ways to grow a rooted tree $(T, \tilde{T})$ starting from the root and adding edges one by one

$$\dots \dots \dots \boxed{\dots} \longleftrightarrow [\hat{X}_u \hat{X}_v, \dots \dots \dots] \dots \dots \dots \boxed{\dots} \longleftrightarrow [\hat{Y}_u \hat{Y}_v, \dots \dots \dots]$$

the number of ways to grow a tree

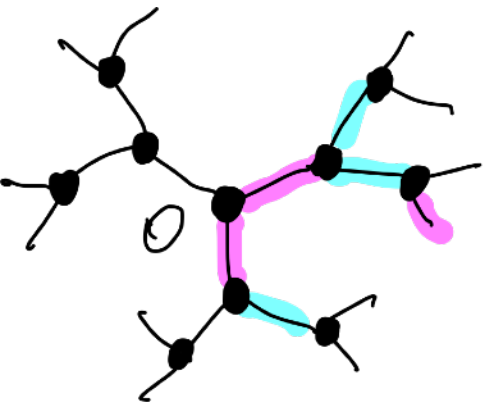
$N(T)$ the number of ways to grow a rooted tree $(T, \tilde{T})$ starting from the root and adding edges one by one



Conjecture: there is $C > 0$ such that for any $n = 1, 2, \dots$ there exists a rooted tree $(T, \tilde{T})$ with $N(T) \geq n!/C^n$

expected in $\mathbb{Z}^d$ with $d \geq 2$

easily proved in the Bethe lattice



the total # of ways to grow trees with n edges $= 3 \times 4 \times \dots \times (n+2)$

the number of distinct rooted trees $\leq 3^{2n}$

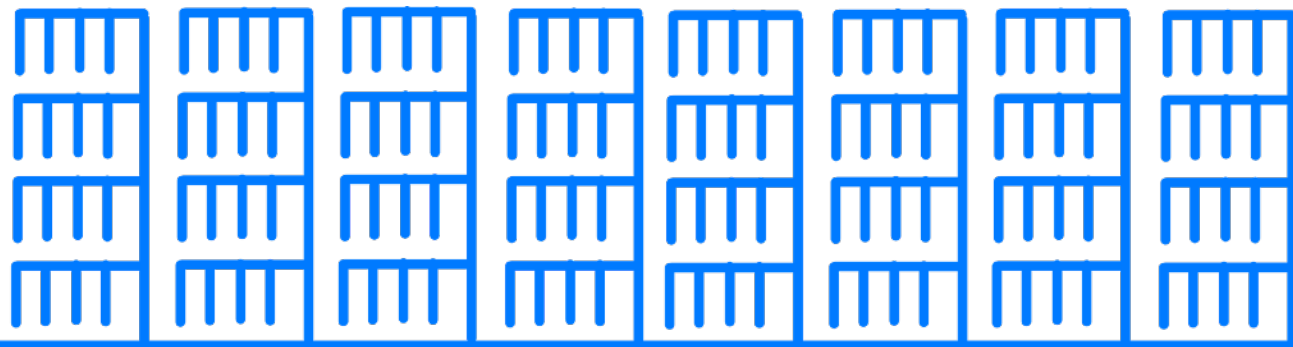
thus $\exists T$ s.t. $N(T) \geq \frac{3 \times 4 \times \dots \times (n+2)}{3^{2n}} \geq \frac{n!}{9^n}$

trees that can be grown in “too many” ways

$N(T)$ the number of ways to grow a rooted tree $(T, \tilde{T})$ starting from the root and adding edges one by one

Theorem (Bouch 2015) there are $C > 0$ and an infinite set $G \subset \mathbb{N}$ such that for any $n \in G$ there exists a rooted tree $(T, \tilde{T})$ with $|T| = n + 1$ and $N(T) \geq n!/C^n$

a tour de force hierarchical construction
see Tasaki (2024)



Bouch tree



Gabriel Bouch

this proves, for $n \in G$, the desired $b_1 b_2 \cdots b_n \geq \alpha^n n!$ for Lanczos coefficients in a large class of models in $d \geq 2$!!

the set G is infinite, but is extremely sparse...

imaginary-time evolution

$$\hat{H} = -\frac{1}{2} \sum_{\substack{u,v \in \mathbb{Z}^d \\ (|u-v|=1)}} \{J_X \hat{X}_u \hat{X}_v + J_Y \hat{Y}_u \hat{Y}_v + J_Z \hat{Z}_u \hat{Z}_v\} - \sum_{u \in \mathbb{Z}^d} \{h_X \hat{X}_u + h_Y \hat{Y}_u + h_Z \hat{Z}_u\}$$
$$S = \frac{1}{2}$$

initial operator $\hat{A} \in \mathfrak{A}_{\text{loc}}, \hat{A}^\dagger = \hat{A}, \|\hat{A}\|_{\text{NHS}} = 1$
(we finally set $\hat{A} = \hat{X}_o$)

$\hat{H}_\Lambda$ the restriction of $\hat{H}$ onto a finite region $\Lambda \subset \mathbb{Z}^d$

$$\hat{A}_\Lambda(i\beta) = e^{-\beta \hat{H}_\Lambda} \hat{A} e^{\beta \hat{H}_\Lambda} = \sum_{m=0}^{\infty} \frac{(i\beta)^m}{m!} \delta_\Lambda^m(\hat{A})$$

$$\beta \in \mathbb{R} \quad \delta_\Lambda(\hat{A}) = i [\hat{H}_\Lambda, \hat{A}]$$

we shall prove $\|\hat{A}_\Lambda(i\beta)\|$ diverges as $\Lambda \uparrow \mathbb{Z}^d$ if $|\beta|$ is large

not straightforward, as the coefficients usually have mixed signs

key observation

$$\hat{A}_\Lambda(i\beta) = e^{-\beta \hat{H}_\Lambda} \hat{A} e^{\beta \hat{H}_\Lambda} = \sum_{m=0}^{\infty} \frac{(i\beta)^m}{m!} \delta_\Lambda^m(\hat{A})$$

$$\delta_\Lambda(\hat{A}) = i [\hat{H}_\Lambda, \hat{A}]$$

$$\langle \hat{A}, \hat{A}_\Lambda(i\beta) \rangle_{\text{NHS}} = \sum_{m=0}^{\infty} \frac{(i\beta)^m}{m!} \langle \hat{A}, \delta_\Lambda^m(\hat{A}) \rangle_{\text{NHS}}$$

noting $\langle \hat{A}, \delta_\Lambda(\hat{B}) \rangle_{\text{NHS}} = -\langle \delta_\Lambda(\hat{A}), \hat{B} \rangle_{\text{NHS}}$, one can show

$$\langle \hat{A}, \delta_\Lambda^m(\hat{A}) \rangle_{\text{NHS}} = \begin{cases} (-1)^{\frac{m}{2}} \langle \delta_\Lambda^{\frac{m}{2}}(\hat{A}), \delta_\Lambda^{\frac{m}{2}}(\hat{A}) \rangle_{\text{NHS}}, & m \text{ even;} \\ 0 & m \text{ odd.} \end{cases}$$

$\langle \hat{C}, \delta_\Lambda(\hat{C}) \rangle = 0$ if $\hat{C} = \hat{C}^\dagger$

$$\langle \hat{A}, \hat{A}_\Lambda(i\beta) \rangle_{\text{NHS}} = \sum_{n=0}^{\infty} \frac{(i\beta)^{2n}}{(2n)!} (-1)^n \langle \delta_\Lambda^n(\hat{A}), \delta_\Lambda^n(\hat{A}) \rangle_{\text{NHS}}$$

$$= \sum_{n=0}^{\infty} \frac{\beta^{2n}}{(2n)!} (\|\delta_\Lambda^n(\hat{A})\|_{\text{NHS}})^2$$

nonnegative for $\beta \in \mathbb{R}$

lower bound for $\|\hat{A}_\Lambda(i\beta)\|$

for finite $\Lambda \subset \mathbb{Z}^d$, take N such that $\delta_\Lambda^n(\hat{A}) = \delta^n(\hat{A})$ for any $n = 1, \dots, N$

$N \uparrow \infty$ as $\Lambda \uparrow \mathbb{Z}^d$

$$\langle \hat{A}, \hat{A}_\Lambda(i\beta) \rangle_{\text{NHS}} = \sum_{n=0}^{\infty} \frac{\beta^{2n}}{(2n)!} \left(\|\delta_\Lambda^n(\hat{A})\|_{\text{NHS}} \right)^2 \geq \sum_{n=0}^N \frac{\beta^{2n}}{(2n)!} \mu_{2n}$$

moment $\mu_{2n} = \left(\|\delta^n(\hat{A})\|_{\text{NHS}} \right)^2$

noting $\langle \hat{A}, \hat{A}_\Lambda(i\beta) \rangle_{\text{NHS}} \leq \boxed{\|\hat{A}\|_{\text{NHS}}} \overset{=1}{\| \hat{A}_\Lambda(i\beta) \|_{\text{NHS}}} \leq \|\hat{A}_\Lambda(i\beta)\|$

we get $\|\hat{A}_\Lambda(i\beta)\| \geq \sum_{n=0}^N \frac{\beta^{2n}}{(2n)!} \mu_{2n}$

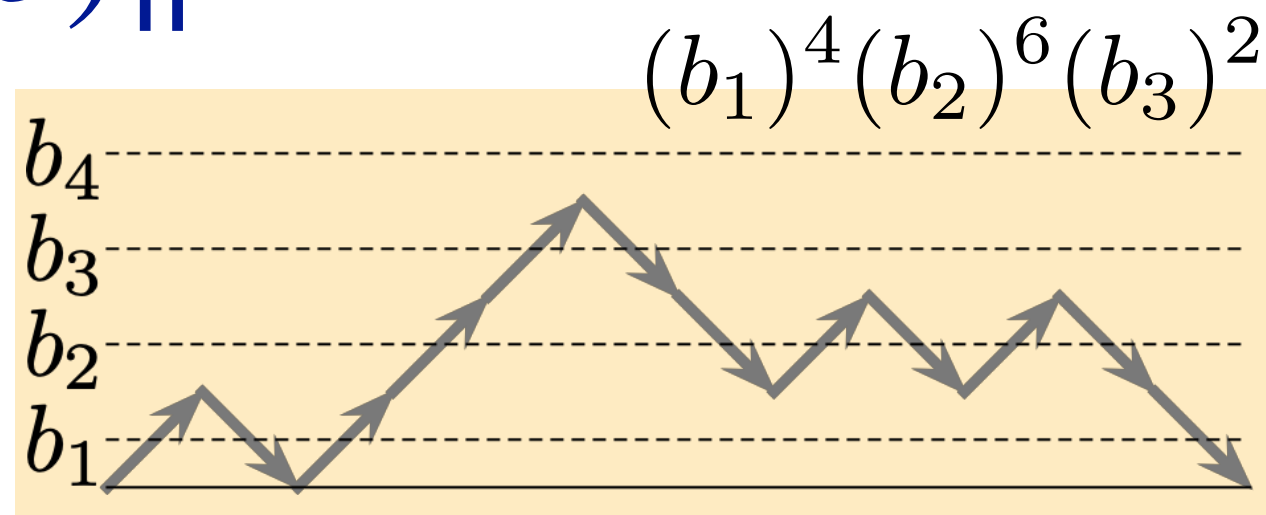
good news!

moments are related to the Lanczos coefficients!

divergence of $\|\hat{A}_\Lambda(i\beta)\|$

$$\mu_{2n} = \sum_{(p_1, \dots, p_{2n})} b_{p_1} b_{p_2} \cdots b_{p_{2n}}$$

sum over Dyck paths



for example $\mu_2 = (b_1)^2$ $\mu_4 = (b_1 b_2)^2 + (b_1)^4$

in particular we see $\mu_{2n} \geq (b_1 b_2 \cdots b_n)^2$

$$\|\hat{A}_\Lambda(i\beta)\| \geq \sum_{n=0}^N \frac{\beta^{2n}}{(2n)!} \mu_{2n} \geq \sum_{n=0}^N \frac{\beta^{2n}}{(2n)!} (b_1 b_2 \cdots b_n)^2$$

$$\geq \sum_{n \in G \cap [0, N]} \frac{\beta^{2n}}{(2n)!} (\alpha n!)^2 = \sum_{n \in G \cap [0, N]} (\alpha \beta)^{2n} \frac{(n!)^2}{(2n)!}$$

we chose $\hat{A} = \hat{X}_o$

diverges as $N \uparrow \infty$ if $\alpha |\beta| > 2$

works for any model where we have $b_1 b_2 \cdots b_n \geq \alpha^n n!$

motivation
definitions and main results
main ideas of the proof
summary and discussion



summary

$$\hat{H} = -\frac{1}{2} \sum_{\substack{u,v \in \mathbb{Z}^d \\ (|u-v|=1)}} \{J_X \hat{X}_u \hat{X}_v + J_Y \hat{Y}_u \hat{Y}_v + J_Z \hat{Z}_u \hat{Z}_v\} - \sum_{u \in \mathbb{Z}^d} \{h_X \hat{X}_u + h_Y \hat{Y}_u + h_Z \hat{Z}_u\}$$
$$S = \frac{1}{2}$$

in all models with $d \geq 2$ except for the classical Ising model, we proved that

✓ the imaginary-time evolution of a local operator exhibits a singularity at a finite “time”

✓ the Lanczos coefficients b_n are bounded as $b_1 b_2 \cdots b_n \geq \alpha^n n!$ for $n \in G$

quantum spin models in $d \geq 2$ generally exhibit chaotic behavior, and we can partially prove it!



future issues

- ☑ prove the existence of trees that can be grown in “too many” ways for any number n of bonds
- ☑ prove a lower bound for the Lanczos coefficient b_n itself rather than the product $b_1 b_2 \cdots b_n$
- ☑ extend Shiraishi’s method to exclude the possibility of quasi-local conserved quantities
- ☑ prove some dynamical properties which are directly relevant to quantum chaotic behavior



The emperor of the South Sea was called Shu [Brief], the emperor of the North Sea was called Hu [Sudden], and the emperor of the central region was called Hun-tun [Chaos]. Shu and Hu from time to time came together for a meeting in the territory of Hun-tun, and Hun-tun treated them very generously. Shu and Hu discussed how they could repay his kindness. "All men," they said, "have seven openings so they can see, hear, eat, and breathe. But Hun-tun alone doesn't have any. Let's try boring him some!"

Every day they bored another hole, and on the seventh day Hun-tun died.

"Zhuangzi", Chapter 7 Fit for Emperors and Kings

<https://terebess.hu/english/chuangtzu.html>

so far, we have bored only three (not very critical) holes

linear growth of b_n

singularity at imaginary time

no conserved quantities

